



URL Shortener

Project - Tanglish Explanation

Namba project enna panradhu - full explanation with code details

Project:	URL Shortener
Stack:	Spring Boot 3 · React 18 · MySQL 8
Language:	Tanglish (Tamil in English Letters)
Date:	05 June 2026

1. Project Enna Panradhu? (Overview)

Namba URL Shortener project onnu simple idea la iruku. Konjam paarkalaa — romba neelamana link irunthal (eg: `https://www.example.com/category/product/details?id=123456`), adhai "`http://localhost:8080/aB3kZ9`" maadhiri short pannurom. Ippo aa short link-la click pannaa, automatically original URL-ku redirect aagidum. Plus, yaar yeppo click pannanga nu analytics also show panrom!

Sura Summary

Long URL ! Short Code generate pann ! Short URL share pann ! Click count track pann ! Analytics chart kaatu

% Enna features iruku?

- Edhanavadhu long URL-ai 6 character short code-aa maathurom (eg: `aB3kZ9`)
- Short URL click pannaa, original page-ku automatic redirect aagidum (302)
- Oru URL-ai yaar yeppo click pannanga nu database-la save aagidum
- Per-URL analytics — daily click chart (last 30 days)
- Global statistics — platform-wide 14-day trend chart
- Paginated URL list — 10 URLs per page
- Copy to clipboard, direct open button — easy UX
- Swagger API docs — `/swagger-ui.html` la full API documentation

2. Enna Technology Use Panninom? (Tech Stack)

% Backend — Server side

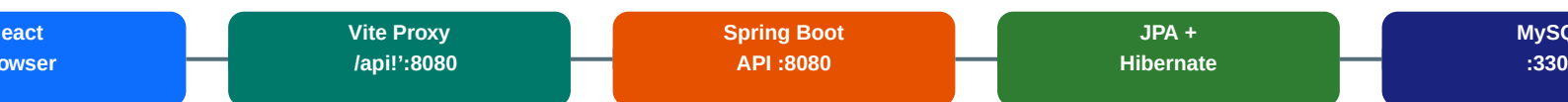
- Java 21 + Spring Boot 3.2 ! Namba server-side logic ellam idhuladhaan iruku
- Spring Data JPA + Hibernate ! Java objects-ai MySQL tables-ku map pannum ORM
- MySQL 8 ! Namba database. URLs, click events ellam idhuladhaan store aagidum
- Lombok ! Getter, Setter, Builder ellam auto-generate pannum. Code shorter aagidum
- SpringDoc ! API documentation auto-generate pannum (Swagger UI)
- Maven ! Build tool. `pom.xml` la dependencies manage panrom

% Frontend — Browser side

- React 18 + TypeScript ! UI components. Type-safe code
- Vite 5 ! Super fast development server and build tool
- Axios ! Backend API-ai call pannum HTTP client
- Recharts ! Interactive charts. `AreaChart`, `ComposedChart` use panni
- Vanilla CSS ! Custom styling. No Bootstrap, no Tailwind

3. System Epdi Work Panradhu? (Architecture)

Namba project 3 layers la work pannum — Frontend (browser), Backend (server), Database. Ipd flow aagidum:



URL shortcuts in a group of thumbnails

- 1. User browser will open URL type manually

- 2. He roBanner.tsx component — UR

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4. Database Epdi Irukkudhu? (Database Design)

Rendu tables iruku namba database-la:

% Table 1: urls — Short URL details store pannum

Column	Type	Enna Pannum?
id	BIGINT (Auto)	Unique number — auto generate
original_url	TEXT	User kotta long URL
short_code	VARCHAR(20)	6-char random code (eg: aB3kZ9)
created_at	TIMESTAMP	Eppo create aacchu
click_count	INT DEFAULT 0	Total clicks — eppo click pannaalum +1
last_accessed_at	TIMESTAMP NULL	Kaadhala click eppo nadanthadhu

% Table 2: click_events — Oru click = oru row

Column	Type	Enna Pannum?
id	BIGINT (Auto)	Unique click ID
url_id	BIGINT (FK)	Edha URL click aacchu nu reference
clicked_at	TIMESTAMP	Exact time of click

Yean rendu tables?

click_count mattum store pannalaam — aana daily breakdown venumna click_events table vendum. DATE(clicked_at) use panni SQL group panni daily stats kaatrom. Ithuthan smart design!

5. Backend Code Epdi Work Panradhu?

%_ [UrlServiceImpl.java](#) — Main Brain

Idhu namba project-la miga important file. Business logic ellam idhuladhaan iruku.

```
// Short code generate pannum method:
private static final String ALPHABET = "ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz0123456789";
private static final int CODE_LENGTH = 6;

// 62 characters ^ 6 = 56 billion possible combinations!
// SecureRandom use pannom – security important

private String randomCode() {
    StringBuilder sb = new StringBuilder(6);
    for (int i = 0; i < 6; i++)
        sb.append(ALPHABET.charAt(RANDOM.nextInt(62)));
    return sb.toString(); // eg: "aB3kZ9"
}
```

%_ [RedirectController.java](#) — Click Track + Redirect

```
@GetMapping("/{shortCode}")
public ResponseEntity<Void> redirect(@PathVariable String shortCode) {
    // 1. Database-la shortCode search pannum
    // 2. click_count + 1 pannum
    // 3. click_events table-la new row insert pannum
    // 4. 302 redirect original URL-ku
    UrlResponse urlResponse = urlService.resolveAndTrack(shortCode);
    headers.add("Location", urlResponse.getOriginalUrl());
    return ResponseEntity.status(302).headers(headers).build();
}
```

%_ [Analytics SQL Query](#)

```
-- Per-URL daily clicks (last 30 days):
SELECT DATE(clicked_at) AS day, COUNT(*) AS clicks
FROM click_events
WHERE url_id = :urlId AND clicked_at >= DATE_SUB(NOW(), INTERVAL 30 DAY)
GROUP BY DATE(clicked_at)
ORDER BY day ASC;

-- Global stats (last 14 days):
SELECT DATE(ce.clicked_at), COUNT(ce.id), COUNT(DISTINCT u.id)
FROM click_events ce JOIN urls u ON ce.url_id = u.id
WHERE ce.clicked_at >= DATE_SUB(NOW(), INTERVAL 30 DAY)
GROUP BY DATE(ce.clicked_at) ORDER BY stat_date DESC LIMIT 14;
```

6. Frontend Epdi Work Panradhu?

%_ App.tsx — Main Component

Idhu namba frontend-la root component. State manage pannum, fetchUrls function irukku, page change handle pannum. Ellaa child components-ukum data pass pannum.

%_ HeroBanner.tsx — URL Input Form

- User URL type pannuvaaru
- Client-side validation — new URL() use panni format check
- POST /api/urls call panni short URL create pannum
- Loading state — "Shortening..." button disabled
- Error message — invalid URL ah, backend error ah show pannum
- Success — onUrlCreated() callback call panni parent update pannum

%_ RecentUrls.tsx — URLs Table

- Paginated URL list table
- Copy button — navigator.clipboard.writeText() use panni copy
- Open button — window.open() use panni new tab la open
- Analytics button — GET /api/urls/{id}/analytics call panni modal open
- Toast notification — "Copied!" message 2.5 seconds show aagidum

%_ AnalyticsModal.tsx — Click Analytics

- Overlay modal — background click pannaa close aagidum
- Short URL info — shortCode, originalUrl, totalClicks show
- AreaChart (Recharts) — daily clicks last 30 days
- Animation — fadeIn + slideUp CSS animation

%_ Statistics.tsx — Global Dashboard

- ComposedChart — Area (clicks) + Bar (creations) combined chart
- GET /api/urls/stats call panni 14-day data fetch
- Date labels formatted as "05th Thu" style
- Sorted oldest to newest (left to right)

7. Project Epdi Run Pannuvathu?

%_ Step 1 — MySQL setup

```
-- MySQL Workbench / command line la run pannunga:
source d:/url-shorterner/database/schema.sql

-- Ippo url_shortener database create aagidum
-- urls table, click_events table create aagidum
```

%_ Step 2 — Password confirm pannu

```
# application.properties — ippo idhaana irukku:
spring.datasource.username=root
spring.datasource.password=root

# Vera password use pannirukaanga? Athai maathunga
```

%_ Step 3 — Backend start pannu

```
# PowerShell la run pannunga:
$env:JAVA_HOME = "C:\Users\udhay\.antigravity-ide\extensions\redhat.java-1.54.0-win32-x64\jre\21.0.10-win32-x64"
$env:PATH = "$env:JAVA_HOME\bin;$env:PATH"
d:\url-shorterner\maven\apache-maven-3.9.6\bin\mvn.cmd spring-boot:run

!' "Started UrlShortenerApplication" nu log varum
!' Backend ready: http://localhost:8080
```

%_ Step 4 — Frontend start pannu

```
cd d:\url-shorterner\frontend
npm run dev

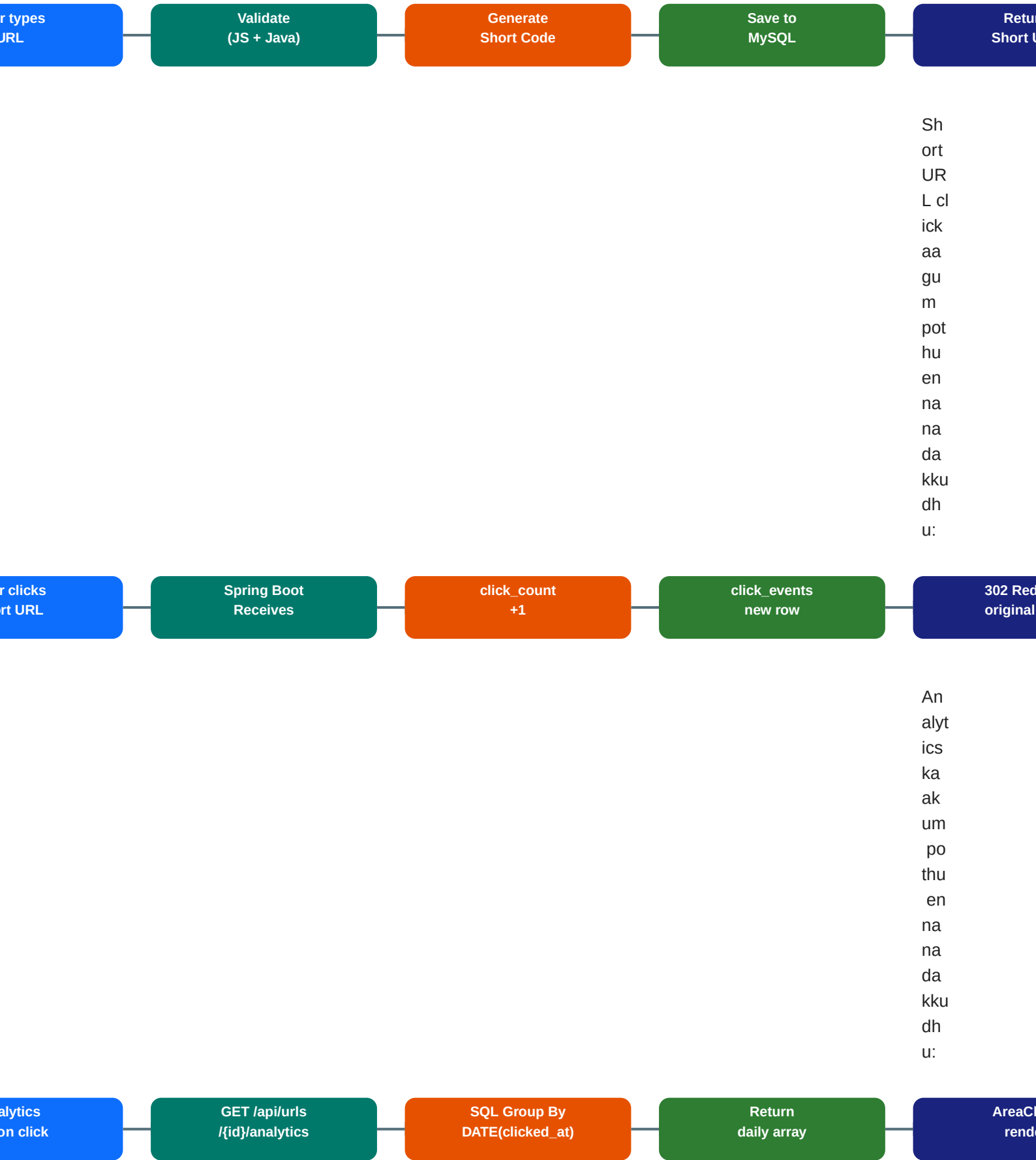
!' Vite ready nu varum: http://localhost:5173
!' Browser la ippa open pannu!
```

Swagger UI — API test pannum tool

<http://localhost:8080/swagger-ui.html> — idha browser la open pannaa, ellaa APIs-iyum directly test panna mudiyum!

8. Full Flow — Visual Summary

URL create aagum pothu enna nadakkudhu:



9. Project Summary — Quick Recap

- Namba project oru complete full-stack web app
- Backend: Java + Spring Boot — REST APIs, business logic, DB access
- Frontend: React + TypeScript — UI, charts, user interaction
- Database: MySQL — persistent storage with 2 related tables
- Analytics: Real-time click tracking with daily breakdown charts
- Yaar URL shorten pannaalum, click panna mudiyum, analytics kaakka mudiyum
- Production-ready features: pagination, error handling, validation, Swagger docs

Oru Line Summary

"Namba project oru URL shortener — long URL-ai short panni, click track panni, analytics kaatrom!"